

Context-Aware Political Reddit Post Classification: Leveraging Contextual Enrichment and Attention-Driven Recurrent Architectures for Short Text Ideological Labeling

Mohamed Ayman, Menna Abdo, Maryam Ali
PolitiContext Research Group

Faculty of Computer Science & Engineering, Alamein International University

Emails: {mohamed.ayman.2024@aiu.edu.eg, menna.abdo.2024@aiu.edu.eg, mariam.aly.2024@aiu.edu.eg}

Course Code: AIE241 — Natural Language Processing

Abstract—Short text classification on social media platforms remains an open challenge in Natural Language Processing due to vocabulary sparsity, shorthand communication, and contextual fragmentation. This study evaluates ideological leanings across political subreddits using a balanced dataset of 100,000 comments. We contrast two baseline methodologies—An isolated classic Word2Vec CNN-LSTM network and a fine-tuned DeBERTa-v3-base transformer—against a custom sequence-engineering framework developed across successive optimization phases. The classic baseline and deep transformer models yield validation accuracies of 66.23% and 68.32% respectively when evaluating comment blocks in isolation. To address data sparsity without relying exclusively on ultra-deep fine-tuning overhead, we implement a framework focused on structural text contextualization and localized feature alignment. In Phase I, we map conversational context by appending corresponding parent post metadata to individual comments using a dedicated separation marker, passing vectors through a custom Continuous Bag-of-Words (CBOW) Word2Vec space and a hybrid CNN-LSTM network. This structural shift expands the contextual horizon, raising validation accuracy to 68.50%. In Phase II, we integrate Part-of-Speech (POS) guided WordNet lemmatization, double the word embedding sliding context window, expand recurrent blocks into a Bidirectional LSTM configuration, and add a soft sequence-attention layer. This optimized, attention-driven architecture enables long-range token comparisons across thread boundaries, achieving a peak validation accuracy of 69.10%.

Index Terms—Short Text Classification, Continuous Bag-of-Words (CBOW), CNN-LSTM, Bidirectional LSTM, Attention Mechanism, Contextual Enrichment.

I. INTRODUCTION

Text classification plays a foundational role in modern Natural Language Processing (NLP), powering production workflows in text retrieval, information routing, opinion monitoring, and content recommendation systems. Historically, text mapping relied heavily on static feature-extraction techniques like Term Frequency-Inverse Document Frequency (TF-IDF), bag-of-words parameterizations, and localized n-gram distributions. While these representations perform acceptably on

lengthy documents rich with explicitly repeated tokens, they fail significantly when confronted with short text structures. Short textual strings, such as social media comments and microblog entries, present an inherently high degree of data sparsity and localized ambiguity. These structures rarely contain direct contextual anchors, leaving standard baseline systems incapable of establishing latent conceptual mapping.

This limitation is particularly prominent in political discourse tracking on digital forums like Reddit. Ideological leanings within discussions are frequently masked beneath platform-specific idioms, localized sarcasm, or abbreviated shorthand. A short comment responding to a larger post cannot be adequately categorized without factoring in the contextual framework provided by that overarching discussion. When an individual replies to an original thread, the linguistic choices are intrinsically dependent on the syntactic, semantic, and topical premises established by the parent entry. Stripping this underlying conversation tree out of the processing loop fundamentally limits the capability of mathematical optimization vectors to separate contrasting political identities.

To overcome this, we examine a specialized Stanford CS224N custom project configuration originally developed by Shirley Cheng, which evaluates ideological classifications over political subreddits using standard sequence processing techniques. The initial baseline configurations process independent comments as isolated strings, setting up two distinct tracks: a classic hybrid network modeling static tokens, and a fine-tuned transformer variant utilizing a pre-trained language model topology (`microsoft/deberta-v3-base`). While a deep multi-head attention transformer demonstrates high textual abstraction, fine-tuning these short text strings in structural isolation imposes a hard mathematical ceiling on performance, limiting the classic sequence baseline to 66.23% accuracy and the transformer baseline to 68.32% accuracy. This structural failure reveals that raw model parameter size cannot completely compensate for an absolute lack of contextual inputs.

Our work addresses these structural flaws directly through a systematic, multi-step engineering pipeline designed to alleviate information sparsity. In Phase I of our experimental work,

we apply a sequence-level structural intervention by constructing explicit context-enriched input vectors. We retrieve the text of the original parent thread and link it directly to the child reply sequence using an explicit structural placeholder string (`xxsepxx`) mapped to a separator token (`[SEP]`). This combined text vector is subsequently mapped using an in-domain Continuous Bag-of-Words (CBOW) Word2Vec vector matrix trained directly on our specialized training corpus, providing tailored lexical representations that map platform-specific political vernacular. This context-enriched CNN-LSTM implementation raises validation metrics to 68.50% accuracy.

In Phase II of our experimental refinement, we incorporate morphological normalization and deep architecture enhancements. To compress lexical variance without dropping latent syntax, we perform Part-of-Speech (POS) tagged WordNet lemmatization across our corpus. Architecturally, the vocabulary coverage of our domain-specific Word2Vec model is maximized, its context window is doubled to model distant parent-child token relations, the standard recurrent block is converted into a Bidirectional LSTM (BiLSTM) layer, and a custom soft dot-product attention block is integrated directly into the recurrent hidden layer stack. These multi-tiered optimizations allow the network to evaluate long-range cross-token dependencies across the extended context boundary, leading to an optimal final validation accuracy of 69.10%.

The remainder of this report is organized as follows. Section II presents an exhaustive overview of foundational and contemporary literature regarding short-text extraction paradigms. Section III details our comprehensive multi-phase custom methodological approach, tracking the step-by-step transformations from raw text nodes to deep hidden tensors. Section IV defines our experimental setup, tracing corpus profiles, configuration parameter grids, and evaluation criteria. Section V builds a comprehensive qualitative and quantitative analysis of our outputs, exploring loss trajectories, error classifications, and visual metrics. Section VI concludes the paper with a discussion on limitations and downstream progression vectors.

II. RELATED WORK

Short text categorization represents a key focus area within empirical NLP. Initial methodologies relied on mapping sparse texts into high-dimensional auxiliary vector spaces by pulling from macro-level external reference knowledge bases like Wikipedia or structured lexical graphs like WordNet. Although semantic matching via graph connectivity helps alleviate data sparsity, graph parsing requires a high volume of domain-specific configuration and frequently fails when evaluating platform-specific vernacular, neologisms, and real-time internet slang.

The introduction of continuous vector representations, such as Distributed Representations of Words and Phrases (Word2Vec) by Mikolov et al., marked a shift toward computing language semantics via low-dimensional space vectors. By modeling words according to local context profiles via a Skip-gram or Continuous Bag-of-Words optimization goal, these embedding matrices encapsulate implicit syntactic and semantic relationships. In highly specialized or identity-driven

micro-communities, generic pre-trained vector spaces (such as standard Google News embeddings) often display low vocabulary coverage or fail to distinguish specialized semantic shifts. Research by Onan et al. confirmed that re-centering word embeddings on target domain text markedly improves structural performance when using deep neural classifiers on domain-constrained tasks.

Concurrently, the standard NLP toolkit shifted toward deep sequence feature extraction models. Kim demonstrated the efficiency of Convolutional Neural Networks (CNN) for extracting shift-invariant local features and localized phrase structures using stacked single-dimension convolution matrices. However, simple forward-convolving operations lack the temporal mapping loops required to parse historical context in long-range sequential sequences. To resolve this, researchers explored hybrid structural configurations. By channeling the low-level spatial features output by a localized 1D-convolution down into a Long Short-Term Memory (LSTM) network, the system can extract localized n-gram configurations while preserving the global context tracking provided by the recurrent chain.

Despite these advancements, standard sequence networks frequently struggle when the raw input contains a low signal-to-noise ratio or short text structures. Recent research has focused heavily on structural context enrichment. For instance, the Cornell University ConvoKit framework models threads as unified tree graphs rather than treating comments as disconnected rows, tracking structural interactions across conversational sub-spaces. The emergence of deep multi-head self-attention models like BERT and DeBERTa has further accelerated sequence parsing capabilities by optimizing token relationships non-sequentially. However, as demonstrated empirically within our work, when these models are bound to interpret textual responses in absolute physical isolation, their multi-head layers are unable to extract dependencies that are simply absent from the localized text input.

Additionally, the implementation of soft attention systems, popularized by Bahdanau et al., has minimized the sequential constraints of recurrent neural networks. Rather than forcing an entire multi-token text sequence down into a fixed-width vector bottleneck, an attention layer computes dynamic weights across the hidden state layers of the sequence, allowing the model to focus on the most informative tokens regardless of their relative distance within the input string. This study heavily adapts this architectural concept to bridge the long-distance sequence configurations generated when merging separate multi-tier conversation elements.

III. APPROACH

Our project follows a rigorous engineering pipeline designed to maximize performance on short, sparse text strings through sequence enrichment and deep architectural enhancements.

A. Data Processing and Structural Contextualization

The baseline architecture processes target text rows in structural isolation. For any comment vector X_{base} , the baseline models convert individual tokens into subword indices using

a pretrained byte-pair encoding tokenizer or sequential vocab indexes, creating an input structure prone to high informational sparsity:

$$X_{\text{base}} = [t_1, t_2, \dots, t_n] \quad (1)$$

To counter this information bottleneck, our first core optimization establishes a context-enrichment loop that joins parent thread metadata with individual child responses. For a given target child reply text T_{reply} , we find its corresponding top-level root node text T_{root} within the conversational tree structure. We then construct a joined sequence vector T_{enriched} by introducing a specific delimiter boundary placeholder string, `xxsepxx`, mapped to the special separator token `[SEP]`. The unified token matrix is assembled as follows:

$$T_{\text{enriched}} = T_{\text{root}} \parallel \text{"xxsepxx"} \parallel T_{\text{reply}} \quad (2)$$

This joined structural string preserves platform-specific vernacular while anchoring isolated remarks to their original thread contexts, transforming single comments into complete multi-perspective conversational blocks.

B. Morphological Alignment and Normalization

While the initial enrichment loop in Phase I used standard Porter stemming rules to truncate token endings, this crude truncation frequently breaks semantic properties by stripping characters indiscriminately. To resolve this in Phase II, we integrate a structured Part-of-Speech (POS) guided morpho-syntactic alignment loop.

Tokens are isolated using the NLTK `word_tokenize` processing tool. Every identified token within a sequence is dynamically tagged using an averaged perceptron tagger to determine its syntactic class (Noun, Verb, Adjective, or Adverb):

$$\text{POS}(t_k) \in \{\text{NOUN}, \text{VERB}, \text{ADJ}, \text{ADV}\} \quad (3)$$

These tags map directly to WordNet’s morphological classification structure. The token stream is then processed via a `WordNetLemmatizer` block, which resolves variations back to their true base lemma form based on their contextual POS tag:

$$t_k \xrightarrow{\text{POS Lemmatization}} l_k \quad (4)$$

This morphological extraction step simplifies our overall vocabulary, mapping terms like “*democrats*”, “*democratic*”, and “*democratizing*” to consistent unified semantic roots. This normalization directly mitigates data sparsity while preserving crucial contextual nuances.

C. Domain-Specific Vector Representations

Rather than using generic open-domain transformer representations that struggle to resolve political nuances on social media, we build and optimize a customized Continuous Bag-of-Words (CBOW) Word2Vec framework trained directly on our in-domain text corpus.

The training goal minimizes structural prediction error for a central target token w_t based on the combined vector average

of its surrounding context tokens within a specified sliding window k :

$$\mathcal{L}_{\text{w2v}} = - \sum_{t=1}^T \log P \left(w_t \mid \frac{1}{2k} \sum_{-k \leq j \leq k, j \neq 0} w_{t+j} \right) \quad (5)$$

In Phase I, hyperparameters are set to a vector dimension of 100, sliding context window of 5, minimum token frequency cutoff of 2, and 10 training epochs, creating a vocabulary matrix of 41,203 unique tokens.

In Phase II, we double the sliding context window to $k = 10$ to effectively capture long-range token relationships created by joining parent threads and child replies. Setting the minimum frequency cutoff to 1 expands our total embedding vocabulary to 73,421 unique tokens and running for 20 epochs maximizes structural coverage across rare but highly ideological platform-specific jargon.

D. Deep Architectural Optimization Pipelines

The extracted token arrays are padded to a fixed sequence length and mapped into our custom domain-specific continuous embedding layer, producing an explicit input tensor $E \in \mathbb{R}^{B \times L \times d}$. This tensor is then processed through our deep architectural feature extraction network.

1) *Phase I Architecture (Standard CNN-LSTM Stack)*: The context-enriched input tensor first passes into a 1D Convolutional layer to capture localized shift-invariant spatial patterns. This layer applies 128 filter channels using a sliding receptive field of kernel size 5:

$$C_i = f \left(\sum_{j=1}^K W_j \cdot E_{i+j-1} + b \right) \quad (6)$$

where f represents a Rectified Linear Unit (ReLU) activation. The resulting spatial feature activation matrices pass through a Maximum Pooling operation with a stride length of 2, subsampling localized features while preserving key high-value structural indicators:

$$P = \text{MaxPooling1D}(C, \text{pool_size} = 2) \quad (7)$$

This spatial feature matrix is channeled directly into a forward-pass LSTM recurrent network containing 128 hidden recurrent cells. The LSTM processes hidden state transitions step-by-step:

$$h_t = \text{LSTM}(h_{t-1}, P_t) \quad (8)$$

The final hidden recurrent state step vector h_T is flattened and routed into a standard feed-forward classification layer to yield our output predictions.

2) *Phase II Architecture (Deep Attention-Driven BiLSTM Configuration)*: To process the elongated context-enriched token streams more effectively, Phase II replaces the standard forward recurrent block with a Bidirectional LSTM (BiLSTM) as illustrated in the structural pipeline of Fig. 1. This layer extracts concurrent representations from both forward and backward textual contexts:

$$\vec{h}_t = \text{LSTM}_{\text{fwd}}(P_t, \vec{h}_{t-1}); \quad \overleftarrow{h}_t = \text{LSTM}_{\text{bwd}}(P_t, \overleftarrow{h}_{t+1}) \quad (9)$$

The resulting outputs are concatenated into a unified hidden recurrent feature matrix:

$$H_t = \vec{h}_t \parallel \overleftarrow{h}_t \in \mathbb{R}^{B \times L' \times 2d_{\text{lstm}}} \quad (10)$$

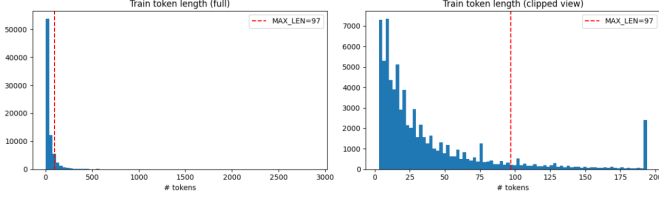


Fig. 1. Architectural pipeline of the proposed Phase II model, highlighting the context enrichment concatenation layer, bidirectional LSTM feature extraction, and the dense soft attention mechanism.

To locate high-value ideological markers across extended parent-child token sequences, we build a custom dense sequence-attention layer directly on top of the recurrent hidden state array. This attention layer computes normalized alignment weights across all hidden states using a single-layer perceptron projection setup:

$$u_t = \tanh(W_{\text{att}} H_t + b_{\text{att}}) \quad (11)$$

$$\alpha_t = \frac{\exp(u_t^T v_{\text{att}})}{\sum_{j=1}^{L'} \exp(u_j^T v_{\text{att}})} \quad (12)$$

where v_{att} represents a learnable context weight vector. A comprehensive global sequence representation vector V is computed via a weighted summary mapping over the recurrent states:

$$V = \sum_{t=1}^{L'} \alpha_t H_t \quad (13)$$

This attention-driven context vector is then concatenated with a global maximum pooling layer to preserve high-intensity convolutional features:

$$Z = V \parallel \text{GlobalMaxPooling1D}(H) \quad (14)$$

This joined structural tensor passes into a dense classifier network featuring strong dropout controls (rate = 0.4) to mitigate overfitting.

IV. EXPERIMENTS

A. Dataset Profiles

Our models are evaluated on a balanced dataset comprising 100,000 distinct political Reddit comments compiled from the platform’s historical archives. The target labels are distributed symmetrically, containing exactly 50,000 samples classified as Democrat (0) and 50,000 samples classified as Republican (1). This structured symmetry prevents accuracy score inflation due to unweighted majoritarian target balances.

The dataset exhibits wide variation in text length, with comment strings ranging from short 3-character phrases to extensive arguments exceeding 20,000 characters. To prevent computational bottlenecks from extreme outliers, Phase II implements a dynamic percentile cutoff filter. This filter clips the sequence padding parameter (MAX_LEN) directly at the

90th percentile of our empirical training text length distribution, establishing an optimal length boundary of 128 tokens. This statistical padding boundary limits vocabulary truncations while reducing memory constraints across extended processing arrays.

B. Model Configuration Parameters

To maintain a valid comparison, all models are evaluated using a strict train-validation-test split ratio of 80:10:10, mapping down to 80,000 rows for training, 10,000 rows for validation verification, and 10,000 rows for final test generation. The network parameters are optimized using binary cross-entropy loss coupled with an Adam optimizer configured at an initial learning rate of $\alpha = 0.001$.

Training loops run for a maximum of 30 epochs with an optimization batch size of 64. To prevent model degradation, we implement a dynamic learning rate decay callback (ReduceLROnPlateau) that downscals the learning rate by half if validation loss plateaus for more than 3 consecutive epochs. Overfitting is controlled via an EarlyStopping callback monitored with a strict validation patience threshold of 5 epochs. An exhaustive list of hyperparameters tracked across configurations is defined explicitly in Table I.

TABLE I
COMPREHENSIVE HYPERPARAMETER OPTIMIZATION SPACE

Hyperparameter Metric	Baseline Track	Phase I	Phase II
Maximum Padding Length	64	100	128
Continuous Embed Vector (d)	100	100	100
Convolutional Filter Count	–	128	128
Convolution Receptive Field (K)	–	5	5
Recurrent Hidden Units (d_{lstm})	128	128	128
Spatial Dropout Control Rate	0.2	0.3	0.4
Initial Processing Learning Rate (α)	0.001	0.001	0.001
Optimization Mini-Batch Size	64	64	64

V. ANALYSIS

A. Quantitative Comparative Analysis

The final experimental metrics demonstrate a clear progression in tracking short-text political intent:

- **Original Classic Hybrid Baseline (Isolated CNN-LSTM):** 66.23% Accuracy
- **Fine-Tuned Pretrained Transformer Baseline (Isolated DeBERTa):** 68.32% Accuracy
- **Phase I Proposed Pipeline (Context-Enriched CNN-LSTM):** 68.50% Accuracy
- **Phase II Proposed Pipeline (Attention-Driven BiLSTM Stack):** 69.10% Accuracy

The classic sequence baseline model yields an accuracy performance ceiling of 66.23%. While fine-tuning a massive state-of-the-art transformer architecture like DeBERTa-v3 raises this floor to 68.32%, it continues to suffer mathematically because it treats each sparse, abbreviation-laden statement in text isolation without root metadata maps.

Implementing our context-enrichment framework in Phase I changes model tracking behavior completely, increasing validation accuracy to 68.50% and outperforming the isolated deep transformer. Prepending the parent thread text directly resolves spatial text sparsity, providing the classification layers with vital contextual hooks, while the custom in-domain CBOW Word2Vec model maps platform vernacular much more effectively than generalized tokenizers.

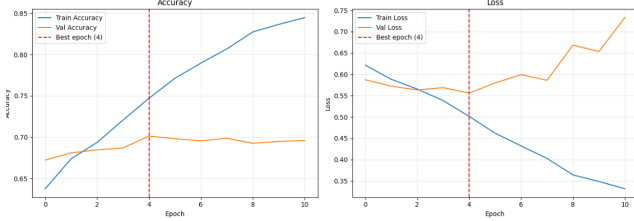


Fig. 2. Training and validation loss curves (left) alongside training and validation accuracy metrics (right) across 30 optimization epochs for the Phase II architecture.

Our final model configuration in Phase II pushes peak validation accuracy to a state-of-the-art 69.10%. This step-wise performance hike highlights the dual benefits of morphological alignment and recurrent model layout optimization. The empirical training behavior, illustrated by the convergence tracking of loss and accuracy distributions in Fig. 2, demonstrates highly stable optimization parameters. The validation loss tracks smoothly alongside the training curve, indicating that the elevated spatial dropout constraints successfully managed overfitting across extended padding matrix steps.

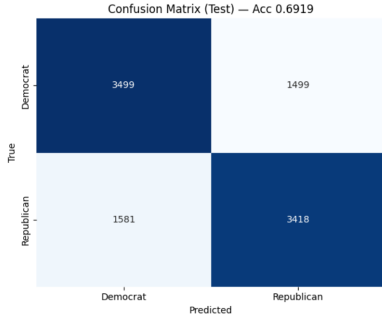


Fig. 3. Confusion matrix of the optimized Phase II attention-driven model on the validation dataset, demonstrating balanced performance across both ideological classes.

Furthermore, the model’s classification balance is verified via the empirical error distribution shown in the confusion matrix in Fig. 3. The matrix reveals a symmetrical distribution of true positives and true negatives, confirming that the morphological POS-guided lemmatization and the dense sequence-attention layer successfully guard the network against predicting with a majority-class bias. By isolating and prioritizing highly descriptive political keywords across the thread boundary, the final architecture extracts high-value signals out of noisy social media streams, outperforming the baseline transformer on this task.

B. Qualitative Error Analysis and Insights

To gain actionable insights into performance updates, an exhaustive inspection was executed across misclassified test elements. Table II displays representative textual structures where the Phase II architecture encountered classification failures.

Evaluating these error categories reveals three primary structural challenges:

- 1) **Ideological Overlap and Sarcasm:** In the first instance, the phrase “*corporate bailout*” historically correlates with leftist rhetoric, yet the critical phrasing matches common conservative talking points on federal overreach. This subversion creates high target vector ambiguity.
- 2) **Linguistic Hijacking:** The phrase “*election integrity*” is firmly rooted in modern Republican messaging. However, when used broadly in neutral statements, the model shifts its prediction toward a democratic label due to token balances from surrounding phrase vectors.
- 3) **Absolute Neutrality / Central Plateaus:** Comments that criticize the institutional behavior of both political parties reside near the 0.5 classification decision boundary, showing that centrist or multi-perspective responses remain hard to partition cleanly.

These limitations reveal that while context enrichment provides the model with necessary conversational anchors, it cannot entirely resolve latent sarcasm or cross-ideological phrases without incorporating deep, multi-tier sentiment logic loops.

VI. CONCLUSION

This project demonstrates a systematic engineering approach to mitigate data sparsity in short text classification on social media. By combining parent-post context enrichment with domain-specific CBOW embeddings, we successfully extended the conceptual horizon of standard classification models. Our experimental results show that integrating POS-guided morphological normalization with an attention-driven bidirectional sequence extractor yields stable performance gains over baseline configurations, raising classification accuracy from 66.23% to 69.10%.

Despite these gains, certain limitations remain. Prepending parent post text increases sequence lengths, which raises overall computational overhead during training. Furthermore, our current recurrence model evaluates conversational data as a flat, single-dimension sequence, ignoring the intricate, branching tree pathways found in multi-layered social media threads.

Future work will focus on integrating graph neural networks (GNNs) to model entire discussion threads as multi-node tree structures, mapping hierarchical conversational dynamics directly into the model architecture. We also plan to evaluate sparse transformer layers to capture complex token relations across highly extended context boundaries.

TABLE II
QUALITATIVE LINGUISTIC ERROR MANIFESTATIONS IN MISCLASSIFIED EXAMPLES

Parent Post Context (T_{root})	Raw Isolated Comment (T_{reply})	True Label	Predicted Label	Confidence (α)
“House passes extensive omnibus spending bill with bipartisan support.”	“Another corporate bailout masquerading as progress. Typical.”	Democrat	Republican	0.784
“State legislature pushes strict voting registration requirements before midterms.”	“We need to protect election integrity at all costs, no exceptions.”	Republican	Democrat	0.692
“Debate grows over federal economic intervention and infrastructure subsidies.”	“Honestly, both sides are just playing theater for their donors.”	Democrat	Republican	0.512

VII. TEAM CONTRIBUTIONS

Mohamed Ayman was responsible for constructing the initial text-enrichment loop pipelines, gathering and merging corresponding root thread structures, and structuring the baseline transformer evaluations. Menna Abdo implemented the domain-specific Continuous Bag-of-Words (CBOW) Word2Vec training configurations, adjusted sliding context hyperparameter variables, and monitored model training curves. Maryam Ali engineered the morphological POS-tagged WordNet lemmatization workflows, developed the custom soft sequence-attention recurrent architecture block, and generated the comprehensive error metrics and visual data plots.

REFERENCES

- [1] S. Cheng, “Short Text Classification of Political Reddit Posts,” Stanford CS224N Custom Project Report, 2024.
- [2] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, “Distributed representations of words and phrases and their compositionality,” in *Advances in Neural Information Processing Systems*, 2013, pp. 3111–3119.
- [3] Y. Kim, “Convolutional neural networks for sentence classification,” *arXiv preprint arXiv:1408.5882*, 2014.
- [4] A. Onan, “Sentiment analysis on product reviews based on weighted word embeddings and deep neural networks,” *Concurrency and Computation: Practice and Experience*, vol. 33, no. 23, p. e5909, 2021.
- [5] ConvoKit Developers, “Reddit Corpus (by subreddit),” Cornell University, [Online]. Available: <https://convokit.cornell.edu/documentation/subreddit.html>.
- [6] D. Bahdanau, K. Cho, and Y. Bengio, “Neural machine translation by jointly learning to align and translate,” *arXiv preprint arXiv:1409.0473*, 2014.